

1 a
$$\binom{n}{k} = \frac{n!}{(n-k)! k!}$$

$$k \cdot \binom{n}{k} = \frac{k \cdot n!}{(n-k)! k!} = \frac{n!}{(n-k)! (k-1)!}$$

$$\& \quad n \cdot \binom{n-1}{k-1} = n \cdot \frac{(n-1)!}{(n-1-(k-1))! (k-1)!}$$

$n-1-k+1 = (n-k)$ can be rewritten
as $\frac{n \cdot (n-1)!}{(n-k)! (k-1)!}$ $n \cdot (n-1)! = n!$

$\frac{n!}{(n-k)! (k-1)!}$ which is equal to the other expression

b The LHS represents the number of ways to choose k books from a shelf with n books $\binom{n}{k}$ and then pick one of those to read (k)
The RHS also represents this, but in the opposite order choosing 1 book from the n on the shelf (n) & then choosing $k-1$ from the remaining to fill out our group of k books.

2 $A = \{1, 2, 3, 4\}$ $B = \{1, 4\}$

Rule 1 $R_1 := \bigvee_{i=1}^4 \bigvee_{j=1}^4 b_{ij}$

at least one cell has a black stone

Rule 2 $R_2 := \left(\bigvee_{i \in B} \bigvee_{j \in B} w_{ij} \right) \wedge \left(\bigvee_{i \in A} \bigvee_{j \in A} \left(\bigvee_{m \in A \setminus \{i\}} \bigvee_{n \in A} w_{ij} \wedge w_{mn} \right) \vee \left(\bigvee_{m \in A} \bigvee_{n \in A \setminus \{j\}} w_{ij} \wedge w_{mn} \right) \right)$

there is a white stone in a corner cell

there are two white cells (i, j) & (m, n) that are not the same (differ in at least either row or column)

Rule 3 $R_3 := \bigwedge_{i \in A} \bigwedge_{j \in A} w_{ij} \Rightarrow \neg b_{ij}$

For all cells on the grid having a white stone means there cannot be a black one (as that would be > 1 per cell)

Rule 4

$$R_4 := \left(\bigwedge_{i=1}^3 \bigwedge_{j=1}^4 b_{ij} \Rightarrow \neg b_{i+1,j} \right) \wedge \left(\bigwedge_{i=1}^4 \bigwedge_{j=1}^3 b_{ij} \Rightarrow \neg b_{i,j+1} \right)$$

No two black stones are vertically or horizontally adjacent

Overall

$$F := \bigwedge_{n=1}^4 R_n \quad \text{All 4 rules must hold at once}$$

3 a Using Contrapositive

$$\frac{x+y}{2} \geq y \Rightarrow x \geq y$$

$$x+y \geq 2y$$

$$x \geq 2y - y$$

$$x \geq (2-1)y$$

$$x \geq y$$

Proved contrapositive, thus original is true $x < y \Rightarrow \frac{x+y}{2} < y$

b Disprove via contradiction

Let $y=1$ & let $x=0$

$\forall z \quad zx = 0$ & $0 < 1$

therefore there does not exist a z for this choice of x, y , so the statement is false

c. Prove via showing internal statement is false (thus its negation is true)

$$\exists x. \forall y. (y \neq x \Rightarrow y < x)$$

This can be shown to be false for all x by considering

$y = x+1$, $y \neq x$ & $y > x$, showing the implication to not hold, no matter the choice

As this statement is false, the original statement, which is its negation is true.

4

the average of (a, b, c) & (d, e, f)

$$\text{is } \left(\frac{a+d}{2}, \frac{b+e}{2}, \frac{c+f}{2} \right)$$

for one of these coordinates to be an integer the two averaged coordinates must both be even or both odd

Both even $k, j \in \mathbb{Z}$

$$\frac{2k + 2j}{2} = k + j \text{ which is } \in \mathbb{Z} \quad \checkmark$$

Both odd

$$\frac{2k+1 + 2j+1}{2} = k+j+1 \text{ which also } \in \mathbb{Z} \quad \checkmark$$

One even one odd

$$\frac{2k + 2j+1}{2} = k+j+\frac{1}{2}, \notin \mathbb{Z}$$

Thus we must have 2 points who have the same pattern of even & odd coordinates

There are $2 \cdot 2 \cdot 2 = 8$ possible patterns for the coordinates these are our holes.

Because we have 9 points (pigeons) The PHP states that at least 2 must have the same pattern of even & odd coordinates & thus this pair's average is on the grid